

Bursting into bloom



ALL PLANTS HAVE SPECIAL CONTROL SYSTEMS that make sure that they flower at the right time. Some need lots of warmth, and others will only flower after the ground has been soaked by rain. But for many, the most important trigger for flowering is the length of the night. Plants cannot see the difference between night and day, but they can sense it with special chemicals that work like a clock. The length of the night alters with the seasons, so this chemical clock keeps a plant on time.

Many plants flower in late spring and early summer, when the days are getting longer and the nights are getting shorter. At this time of the year, the air is often filled with insects, which makes it an ideal time for flowers to be pollinated. Other plants, such as the garden nasturtium, wait until it is midsummer. This gives them plenty of time to grow before their flowering time arrives. A small number of plants, including chrysanthemums, need short days and long nights before they will flower. They bloom in late summer and early fall, when most other plants have finished flowering and have already made their seeds.

New petals
unfolding outward

Petals folded
back

Sepal

Petal

THE FLOWER OPENS

The garden nasturtium belongs to a family of plants that comes from South America. In countries farther from the equator it flowers in midsummer. When the light conditions are right for the plant, flower buds begin to form. Each flower bud is protected by five sepals. As the bud starts to burst, the sepals open to reveal five bright orange petals that grow outward and fold back. One of the sepals develops a long spur that lies at the back of the flower. This spur produces nectar, which attracts insect pollinators to the flower.

BLOOMING LOVELY

Markings called honeyguides show insects the way to the nectar. To reach it, the insects have to clamber over the anthers (p. 17), which dust them with pollen. As the days pass, the anthers wither and the three stigmas (p. 17) become receptive to the pollen of other plants. Insects in search of nectar now dust the stigmas with pollen.